# Andhra University College of Engineering

Civil Engineering | Course Code: CE301

## Structural Analysis - Comprehensive Syllabus Lecture Notes

Syllabus Focus: CE301 Full Syllabus Comprehensive Lecture Notes Units 1-5

Summary: Complete syllabus lecture notes covering Structural Analysis. Includes detailed unit-b

### I. UNIT-BY-UNIT COMPREHENSIVE SYLLABUS CONTENT

#### Unit 1: Energy Theorems & Deflections

" Syllabus Scope: Castigliano first and second theorems, Unit load method, Strain energy in bending and t

#### Unit 2: Slope Deflection Method

" Syllabus Scope: Derivation of slope deflection equations, Analysis of continuous beams with support set

# CE301: Structural Analysis - Advanced Topics & Assessment

Academic Track: Civil Engineering - Units 3, 4, 5 & Examination Matrix

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## II. ADVANCED SYLLABUS UNITS (UNITS 3 TO 5)

**Unit 3: Moment Distribution Method**

" Stiffness and carry-over factors, Distribution factors, Analysis of continuous beams and portal frames witho

**Unit 4: Moving Loads & Influence Line Diagrams**

" Influence lines for reaction, shear force, and bending moment in simply supported girders, Muller-Breslau pr

**Unit 5: Arches & Cable Structures**

" Three-hinged and two-hinged circular and parabolic arches, Normal thrust, Radial shear, Bending moment diagr

## III. HIGH-YIELD SEMESTER EXAM QUESTION BANK & SOLUTIONS

1. Analyze a continuous two-span beam with fixed ends subjected to uniformly distributed loads using Mome
2. Apply Castigliano second theorem to calculate vertical deflection at the loaded node of an indetermina
3. Draw the influence line diagram for bending moment at the center of a 20-meter span girder and find ma

## IV. PRESCRIBED TEXTBOOKS & REPOSITORY CITATIONS

* C.S. Reddy - Basic Structural Analysis Tata McGraw-Hill 3rd Ed
* R.C. Hibbeler - Structural Analysis Pearson 10th Ed

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